QUANTUMFLOOR

Computational Research Report

Job ID: 98c79f65

1. Executive Summary

This report details the execution of the QuantumFloor diagnostic pipeline on dataset 'label'. The system evaluated multiple classical baselines against quantum algorithms to determine the optimal strategy.

2. Dataset Intelligence

Modality:	TABULAR
Total Rows:	80
Unique Rows:	80
Duplicate Rows:	0
Total Columns:	5
Missing Values:	0.00%
Leakage Risk Detected:	0

3. Data Preparation

Train/Test Split:	80/20 Stratified
Preprocessing:	Pipeline (Imputation + Scaling)
Leakage Protection:	Enforced

4. Computational Plan

Original Features:	5
Quantum Embedding Dimension:	2
Qubit Budget:	2
Classical Training Size:	64
Quantum Subset Size:	64

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5. Quantum Execution

Execution Backend: SIMULATED EXECUTION

Ansatz: AngleEmbedding + IsingZZ

Qubits: 2

Circuit Depth: 2

Total Operations: 2

6. Classical vs Quantum Baselines

Model	Type	AUROC	F1	Spec@95	Notes
full_feature_classical	Classical	0.5078	0.5000	0.0000	
logreg	Classical	0.3281	0.5263	0.0000	
svm_rbf	Classical	0.3750	0.3077	0.0000	
random_forest	Classical	0.6875	0.6667	0.5000	
qsvm	Quantum	0.5625	0.4286	0.3750	
pqk	Quantum	0.7500	0.6667	0.6250	
vqc	Quantum	0.3438	0.6087	0.0000	

7. Comparison & Explainability

Best Classical AUROC: 0.6875

Best Quantum AUROC: 0.7500

Delta: 0.0625

Explainability Status: available

PC1: 0.1468

PC2: 0.1290

8. Final Verdict

Best Overall Model: pqk

Quantum Advantage: Yes